

In this assignment a perceptron was implemented to test the linear separability of boolean functions of dimension $n = 2, 3, 4, 5$. For each n the fraction of linearly separable functions given by the algorithm is compared to that of the theoretical value.

For a given boolean function we give all 2^n possible inputs with their corresponding function boolean truth at which the perceptron is trained for 20 epochs with learning rate $\eta = 0.05$ and weights and threshold initialized as $w_i \sim \mathcal{N}(0, \frac{1}{n})$, $i = 1, \dots, n$ and $\theta = 0$ respectively. The perceptron then evaluates if it is possible to find a hyperplane separating the patterns.

For dimension $n = 2, 3$ we look at all possible boolean functions (16 and 256 functions respectively) and the algorithm finds the estimated fraction of linear separability to be 0.87500 for $n = 2$ and 0.40625 for $n = 3$, as can be seen in table 1, which perfectly matches that of the theoretical values.

For $n = 4, 5$ we instead only look at 10000 possible boolean functions, which are sampled randomly from the total pool of functions. Here in table 1 we now observe the estimated average fractions 0.02874 and 0.00004 for $n = 4$ and $n = 5$ respectively. The estimated fractions from the method now no longer perfectly match the theoretical fractions.

n	Theoretical fraction	Average fraction	Standard deviation
2	0.87500	0.87500	0.00000
3	0.40625	0.40625	0.00000
4	0.02872	0.02874	0.00169
5	0.00002	0.00004	0.00008

Table 1: The theoretically known fraction of linearly separable Boolean functions compared with the average fraction and standard deviation obtained from the perceptron simulations. For each n , the method was repeated 20 times. For $n = 2, 3$, all Boolean functions were tested, while for $n = 4, 5$, 10000 Boolean functions were sampled in each repetition. The perceptron was trained for 20 epochs using learning rate $\eta = 0.05$, with weights initialized from a normal distribution with mean zero and variance $\frac{1}{n}$ and threshold initialized to zero.

The exact fractions for $n = 2, 3$ is not surprising, since every boolean function is examined and the dataset of inputs is small (4 and 8 patterns). This along with the 20 epochs gives the perceptron adequate conditions to find potential hyperplanes. For $n = 4, 5$ on the other hand we only consider a fraction of the total possible boolean functions, making perfect alignment with the theoretical ratio impossible. Nevertheless, the estimated averages remain close to the theoretical.

A point of potential inaccuracy is that the functions are sampled with replacement, making it possible for the same function to be considered more than once. However as this effect is averaged out over 20 runs this effect will not be significant.

Overall, the perceptron reliably detects linear separability. Exhaustive evaluation in low dimensions ($n = 2, 3$) over the boolean function space yields the theoretical fractions exactly, while random sampling in higher dimensions ($n = 4, 5$) gives accurate estimates. The results affirm the sharp decline of the fraction of separable boolean functions with dimension.